AI Fake Medical Advice Detector

Advanced AI Healthcare Project

Objective

Build an advanced AI system that detects and classifies fake or misleading medical advice from health-related text.

The system should analyze health advice using medical NLP models and classify whether the advice is medically correct or misinformation.

This project should leverage modern AI tools such as MedGemma, HuggingFace Transformers, and NLP pipelines.

Tools (Free)

Python

HuggingFace Transformers

MedGemma (or open medical LLM)

spaCy / ScispaCy

Google Colab

Optional: Streamlit or Gradio

Project Requirements

The AI system must:

1. Accept health advice text as input.

2. Detect medical entities and claims.

3. Analyze whether the advice aligns with reliable medical knowledge.

4. Classify the advice as:

   - Medically Correct

   - Misinformation

5. Provide a short explanation for the classification.

Example

Input:

"Drinking lemon water cures cancer completely."

Output:

Classification: Misinformation

Explanation: There is no scientific evidence that lemon water cures cancer.

Implementation Steps

Step 1: Environment Setup

Install required libraries:

pip install transformers

pip install torch

pip install spacy

pip install scispacy

Step 2: Data Collection

Collect medical advice examples from:

- WHO health guidelines
- CDC medical resources
- Kaggle health datasets

Create a dataset labeled as:

- correct advice
- misinformation

Step 3: Text Preprocessing

Perform:

- tokenization

- stopword removal

- normalization

Step 4: Medical Entity Recognition

Use spaCy or ScispaCy to detect medical entities such as diseases, symptoms, and treatments.

Step 5: Model Integration

Load MedGemma or a medical NLP model from HuggingFace.

Step 6: Classification Pipeline

Create a classification pipeline that evaluates whether the claim is supported by medical knowledge.

Step 7: Output Explanation

Use the LLM to generate a short explanation describing why the advice is correct or misinformation.

Advanced Features (Bonus)

Implement at least one advanced feature:

1. Source Verification

Check whether the advice appears in trusted medical sources.

2. Severity Detection

Detect harmful advice that could endanger patient health.

3. Confidence Score

Provide probability score for prediction.

4. RAG-based Knowledge Check

Use a small knowledge base of medical articles to verify claims.

Deliverables

1. Python notebook (Google Colab or Jupyter)

2. GitHub repository

3. Sample inputs and outputs

4. Architecture diagram

5. Short documentation